Supporting Information for

Developmental asymmetries in equilibrium selection

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1. Experimental design and instructions

A. Overview of studies. The paper draws on two studies that share the same experimental paradigm: two one-shot coordination games followed by a stated justification for each choice. The two studies share the same game design and instructions (see Section B) but differ in setting and population. Study 1 is primarily a lab-in-the-field classroom study and spans ages 10–20 through participants in late primary and secondary education, vocational training, and a small university pilot. Study 2 consists of laboratory sessions with high-school and university participants aged 17–20.

Study 1 — Lab-in-the-field, TeensLab. Study 1 constitutes a subsample of the TeensLab dataset (see (1) for details), a large-scale initiative by a consortium of Spanish universities dedicated to behavioral research with adolescents, whose data are publicly available in https://zenodo.org/records/13720112. Rather than inviting participants to a lab, we took the experiment to them. A total of 1716 students participated in this study. Between January 2023 and May 2023, we visited 20 schools across two Spanish provinces — Barcelona (19 schools, 91.2% of the sample) and Cádiz (1 school, 8.8%) — and ran sessions directly inside the classrooms, with students sitting at their usual desks.

The participant pool was recruited through agreements with school headmasters, who agreed to integrate the experiment into their pedagogical curriculum and to carry it out as a classroom activity. The experiment was approved by the Ethical Committee of Universidad Loyola Andalucía (No. 20190318, 20200709 and 20230301) under an opt-out IRB consent procedure. Because the activity was part of the schools' curriculum, parental consent for participants under 16 was covered by the consent parents had previously given to the school for curricular activities. All participants gave informed consent on an initial screen before starting the questionnaire.

The school-based sample ranged from grades 5 to 11 (ages 10–17), spanning late primary, compulsory secondary education, and high school. Students in vocational training programs were also recruited, reaching a total of 111 students. Additionally, 58 undergraduate students from Universidad Loyola Andalucía participated in the study as a pilot representing 3.38% of the Study 1 total sample. These university participants were aged up to 20 years old. The experiment was conducted as a regular class activity using a self-administered questionnaire programmed in a tailored online platform named SAND.* Subjects answered on their computers, tablets, or mobile phones with guaranteed anonymity. Throughout the main sample, payoffs were hypothetical — a deliberate choice justified by evidence that adolescents do not alter their behavior when real money is introduced (2).

Study 2 — Laboratory experiments. The data for Study 2 are drawn from three sessions of a laboratory experiment described in Brañas-Garza et al. (3). All were conducted at Universidad Loyola Andalucía in designated classrooms, where participants read the instructions and made their decisions using their own mobile phones via a Qualtrics survey. Participants gave informed consent on an initial screen before starting the experiment, under the approval of the Ethical Committee of Universidad Loyola Andalucía (No. 20230301). Incentives were randomized at the individual level ($p = 1/2$): each participant was assigned to either a monetary or a hypothetical payment condition. Participants assigned to the hypothetical condition were not left empty-handed: the Theory of Mind task was incentivized for them, ensuring every participant had a realistic chance of earning money regardless of their assigned condition.

Session 1 consisted of high-school students ($n = 137$) attending the university as part of school-organized whole-class visits, and Session 2 consisted of Loyola students ($n = 105$). Games were presented in a fixed order (Game 1 $\rightarrow$ Game 2). Average earnings were €2.19, including a €1 show-up fee, for a session lasting under 10 minutes. Sessions 1 and 2 were conducted in November 2023 and November 2024, respectively. Session 3 was composed of Loyola students ($n = 301$), in which the order of the games was randomized at the individual level. In Session 3, both the incentive scheme and the game sequence were randomized at the individual level, allowing us to assess whether either design feature affected outcomes. Average earnings were €1.98, again for under 10 minutes of participation. Brañas-Garza et al. (3) show that neither game order nor the incentives affect outcomes. Session 3 was conducted in December 2024.

For the purposes of this paper, we retain participants aged 17–20, excluding those above age 20 due to small cell sizes, as well as a small number of participants misclassified or enrolled as grade repeaters in secondary education. This yields a pooled Study 2 sample of $n = 415$.

B. Design. Both games are one-shot, 2×2 coordination games with two pure-strategy Nash equilibria. Participants were told that they would be paired with a random partner and that they could not communicate with that person. The pairing pool consisted of classmates from the same classroom context in Study 1 and participants from the same experimental session in Study 2. Both players faced the same game structure and the same payoff matrix. The games differ in which feature of the payoff matrix is relevant for equilibrium selection: Game 1 has unequal equilibrium payoffs, whereas in Game 2 the distinction between the equilibria appears under miscoordination. The payoff values below correspond to Study 1; in Study 2 the same amounts were divided by 10, which leaves all the equilibrium structure intact.

Game 1 — Payoff-dominant coordination game. Players choose between Yellow and Red. Pressing Yellow when the other person also presses Yellow gives both players 10 euros; pressing Red when the other also presses Red gives both 5 euros; and any mismatch leaves both with nothing. See payoff matrix in Table S1.

*SAND (Social Analysis and Network Data; https://sand.kampal.com) is a tailored online platform designed for data collection in educational settings, enabling privacy-preserving management of large simultaneous classroom sessions.

Table S1. Game 1 — Payoff-dominant coordination game.

	Yellow (A)	Red (B)
Yellow (A)	**10, 10**	0, 0
Red (B)	0, 0	5, 5

Yellow/Yellow Pareto-dominates Red/Red, making Yellow the payoff-dominant action.

After each game, participants were asked to choose one of four predefined reasons for their decision, presented in random order. For Game 1:

(a) Because I earned more;

(b) Because I earned less;

(c) Because I liked the color;

(d) I chose one randomly.

We define *PayD* as the indicator for pressing Yellow and reporting "*Because I earned more*". PayD therefore identifies the payoff-dominant action accompanied by the congruent stated reason; because the reason is elicited after the choice from a closed list, it does not establish the mental process that generated the choice.

Game 2 — Risk-dominant coordination game. Players choose between White and Blue. White/White and Blue/Blue both pay $(10, 10)$, so successful-coordination payoffs do not distinguish between the two equilibria. The difference appears under miscoordination: pressing White when the other presses Blue gives you 5 euros and them nothing; pressing Blue when the other presses White gives you nothing and them 5 euros. White is therefore the risk-dominant choice. See payoff matrix of Game 2 in Table S2.

Table S2. Game 2 — Risk-dominant coordination game.

	White (A)	Blue (B)
White (A)	**10, 10**	5, 0
Blue (B)	0, 5	10, 10

For Game 2, the four predefined reasons were:

(a) Because it was less risky;

(b) Because it was more risky;

(c) Because I liked the color;

(d) I chose one randomly.

We define *RiskD* as the indicator for pressing White and reporting "*Because it was less risky*". RiskD therefore identifies the risk-dominant action accompanied by the congruent stated reason; as with PayD, it does not establish the mental process that generated the choice.

The full experimental sequence was Game 1 → Explanation 1 → Game 2 → Explanation 2 (with game order randomized only in session 3 of Study 2, and with the belief block described below preceding or following this sequence in the six schools where it was administered).

Belief elicitation. In six of the twenty schools of Study 1 ($n = 403$, ages 12–15 and 18–20), the questionnaire included a further block in which participants reported, for each game, which button they believed their partner would press and which of the same four reasons they attributed to the partner. The belief items mirrored the own-choice items in wording and response options, phrased in the third person. The order of the two blocks—own choice and reason first, or partner belief first—was randomized at the participant level, so that roughly half of participants stated their belief before making their own decision (Game 1: 213 of 403; Game 2: 197 of 396). Within each block the sequence was always action followed by reason.

C. Instructions. Below we reproduce the exact text shown to participants on each of the six screens. The wording is identical across both studies; the amounts shown in the instructions were divided by 10 for Study 2, leaving all the equilibrium structure intact. In Study 1, the screens appeared in Spanish; the following is a faithful English translation. Button colors were not randomized: the equilibrium options were always Yellow for Game 1 and White for Game 2, ensuring the salience structure was consistent across participants. Button positions — left or right — were randomized independently for each participant to prevent any systematic positional bias.

Screen 1 (Introduction):

You will now be paired with a random partner.

Both of you will need to press a button without being able to communicate with each other.

Depending on what each of you chooses, you will receive different (hypothetical) rewards.

[Click the forward button to move on to the first question.]

Screen 2 (Game 1 — payoff matrix):

These are the possible outcomes:

If you press yellow and she/he presses yellow, you both win 10 euros each.

If you press yellow and she/he presses red, you both win 0 euros each.

If you press red and she/he presses yellow, you both win 0 euros each.

If you press red and she/he presses red, you both win 5 euros each.

Which button do you press?





Screen 3 (Explanation 1):

Why did you choose the previous button?

○ Because I earned more
○ Because I earned less
○ Because I liked the color
○ I chose one randomly

Screen 4 (Transition):

Now you are going to play again with another person, other buttons, and other prizes.

Again, you have to choose a button without knowing what the other one does.

Screen 5 (Game 2 — payoff matrix):

These are the possible outcomes:

If you press white and she/he presses white, you both win 10 euros each.

If you press white and she/he presses blue, you will earn 5 euros and she/he will win 0 euros.

If you press blue and she/he presses white, you will earn 0 euros and she/he will win 5 euros.

If you press blue and she/he presses blue, you both win 10 euros each.

Which button do you press?





Screen 6 (Explanation 2):

Why did you choose the previous button?

○ Because it was less risky

○ Because it was more risky

○ Because I liked the color

○ I chose one randomly

D. Tasks. Beyond the two coordination games, participants completed a short battery of tasks designed to capture cognitive abilities. The battery was not identical across studies — some tasks require a minimum level of abstraction that makes them unsuitable for 10-year-olds, and others are redundant once we have university students. Table S3 summarizes which tasks appeared where.

Table S3. Tasks administered in each study.

Task	Study 1 (TeensLab)	Study 2 (Qualtrics)
Cognitive Reflection Test (CRT)	Yes	Yes
Financial Abilities (FinAb)	Yes	—
Probabilistic Reasoning (Del)	Yes	—

Cognitive Reflection Test (CRT). The CRT was administered in both studies. We used the same task as in Alfonso et al. (2), based on Thomson et al.'s (4) task, which preserves the spirit of Frederick's (5) original three-item test while reducing reliance on domain-specific vocabulary that could disadvantage younger or less math-oriented respondents. Each item has a tempting intuitive answer that is wrong, and the correct answer requires a moment of deliberate reflection. Items were randomized in order. The score runs from 0 (all intuitive) to 3 (all reflective). The three items are reproduced below.

- **Q1.** Emilia's father has three daughters. The first two are named April and May. What is the name of the third daughter? (Reflective: **Emilia**; Intuitive: June)
- **Q2.** If you are running a race and you overtake the person in second place, what position are you in? Indicate with a number (e.g., 1 = first, 2 = second, etc.). (Reflective: **2**; Intuitive: 1)
- **Q3.** In a library, the number of books doubles every month. If it takes 48 months for the library to become full, how many months would it take for it to be half full? Indicate with a number. (Reflective: **47**; Intuitive: 24)

Financial Abilities (FinAb). In Study 1, we complemented the CRT with a short financial literacy task.[†] The three items require basic arithmetic with money and an understanding of how interest compounds over time. Scores run from 0 to 3.

- **Q1.** If there are 5 people who own the winning lottery ticket and the prize to be shared is 2 million euros, how much money will each person receive? Write the amount using numbers only, without dots or commas. (Correct answer: **400,000**)
- **Q2.** Imagine you have 100 euros in a savings account. The account earns an interest rate of 10% per year. How much money will you have in the account after 2 years? (Correct answer: **121 euros**)
- **Q3.** Imagine you have 100 euros in a savings account and the interest rate you earn is 2% per year. If you keep the money in the account for 5 years, how much money will you have at the end of those 5 years?
  - Less than 102 euros
  - 102 euros exactly
  - **More than 102 euros** (Correct answer)
  - I don't know

Probabilistic Reasoning (Delavande test). Three probabilistic reasoning items (*Del*) adapted from Delavande and Rohwedder (6) were administered exclusively in Study 1. Participants received one point for each item in which they correctly respected a basic probabilistic relation, for a total score ranging from 0 to 3. The three scored comparisons were: (i) stating that the probability of drawing the green apple from a basket of 5 (1 green, 4 red) is higher than from a basket of 10 (1 green, 9 red); (ii) stating that the probability of eating rice in the next month is higher than in the next week; and (iii) stating that the probability of showering at least once in the next month is higher than the probability of not attending school for a single day in the same period. Violations occur when the reported probability for the broader or more likely event is numerically lower than that for the narrower or less likely event.

[†] Items are drawn from Alfonso et al. (2). They span basic arithmetic with money and simple compound interest, and were calibrated so as not to require knowledge beyond the primary school curriculum.

Participants responded by entering a number from 0 to 100 (50% of participants) or by moving a slider on a 0–100 scale (50% of participants); randomization was at the individual level. The instructions and questions were identical across both response formats. Questions (Q1–Q6) are shown below.

Screen 1 — Instructions

In the next screen, we **will ask you** about the **probability** that certain things could happen. In each question, you should **write down** how **likely you think** it is that these things will happen in reality.

If you think something will happen for **sure**, you should write **100**. If you are fairly but not completely sure, write a number close to 100 (but less than 100). If you think it is something that will **never** happen, write **0**. If you think it is unlikely, write a number close to 0 (but greater than 0). If you think it is **equally likely** to happen or not to happen, write **50**.

Screen 2 — Examples

EXAMPLES:

- If it is a sunny day and I ask you how likely it is that it will be sunny within 1 hour, you would typically say 100 or numbers very close to 100.
- If it is 12 o'clock at night, the most reasonable thing to say is that the probability that it will be sunny in 1 hour is 0.
- If I flip a coin, the probability that it will come up heads is 50.

***Del* questions (2 by screen)**

Q1. Imagine that I have a basket with 5 apples: 1 green and 4 red. If I ask you to choose one of the apples without looking inside the basket, how likely is it that you pick the green apple, from 0 to 100?

Q2. Imagine that I have a basket with 10 apples: 1 green and 9 red. If I ask you to choose one of the apples without looking inside the basket, how likely is it that you pick the green apple, from 0 to 100?

Q3. How likely is it that you eat rice in the next week from 0 to 100? (Including today)

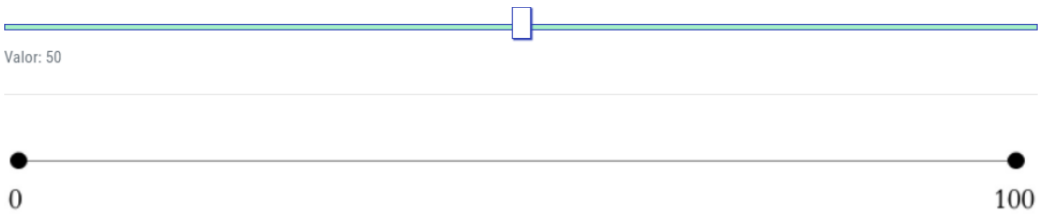
Q4. How likely is it that you eat rice in the next month from 0 to 100? (Including today)

Q5. How likely is it that you do not attend your educational center even once in the next month from 0 to 100? (Including today)

Q6. How likely is it that you take a shower at least once during the next month from 0 to 100? (Including today)

The slider version (used by 50% of participants, randomized at the individual level) presented the same questions and instructions, with participants dragging a marker along a 0–100 scale ("Valor: 50" shown by default) instead of typing a number. See Figure S1 for an illustration of the slider interface.

Fig. S1. Slider interface



Note: Screenshot of the slider response format used by half of the participants in the probabilistic reasoning task. Participants dragged a marker along a 0–100 scale; the default position was 50 ("Valor: 50"). The other half of participants typed a number directly.

2. Sample composition

This appendix provides a full breakdown of participants across both studies. The samples span a wide range of educational stages, from late primary school to university, generating substantial variation in age, cognitive development, and educational experience. Each study's sample is described along two complementary dimensions. First, we report age in years (Tables S4 and S6), which provides a continuous developmental scale that facilitates comparison across cohorts and studies. Second, we report participants' academic track or grade (Tables S5 and S7), which reflects the structure of the educational system used for recruitment. In the Spanish system, grades 5–6 correspond to late primary education, grades 7–10 to compulsory secondary education, and grades 11–12 to academic high school. Vocational training (VT) and university (U) students are reported as separate categories, as they represent distinct educational pathways relative to the standard academic track. Table S8 presents the distribution of our full sample by age in years.

Table S4. Study 1 — Sample distribution by age.

	10	11	12	13	14	15	16	17	18	19	20	Total
N	136	208	237	276	300	244	124	24	78	53	36	**1,716**
% of sample	7.9%	12.1%	13.8%	16.1%	17.5%	14.2%	7.2%	1.4%	4.6%	3.1%	2.1%	**100.0%**
% Female	53.7%	51.4%	43.5%	46.0%	47.0%	52.5%	43.5%	50.0%	60.3%	43.4%	55.6%	**48.7%**

Table S5. Study 1 — Sample distribution by grade.

	5th	6th	7th	8th	9th	10th	11th	12th	VT	U	Total
N	173	213	259	299	340	190	73	–	111	58	**1,716**
% of sample	10.1%	12.4%	15.1%	17.4%	19.8%	11.1%	4.2%	–	6.5%	3.4%	**100.0%**
% Female	53.8%	47.4%	47.5%	45.8%	45.6%	53.2%	46.6%	–	70.3%	22.4%	**48.7%**

Table S6. Study 2 — Sample distribution by age.

	17	18	19	20	Total
N	119	79	144	73	**415**
% of sample	28.7%	19.0%	34.7%	17.6%	**100.0%**
% Female	54.6%	40.5%	47.2%	46.3%	**46.3%**

Table S7. Study 2 — Sample distribution by grade.

	12th	University	Total
N	119	296	**415**
% of sample	28.7%	71.3%	**100.0%**
% Female	54.6%	42.9%	**46.3%**

Table S8. Full sample — All studies combined, by age.

	10	11	12	13	14	15	16	17	18	19	20	Total
N	136	208	237	276	300	244	124	143	157	197	109	**2,131**
% of sample	6.4%	9.8%	11.1%	12.9%	14.1%	11.4%	5.8%	6.7%	7.4%	9.2%	5.1%	**100.0%**
% Female	53.7%	51.4%	43.5%	46.0%	47.0%	52.5%	43.5%	53.8%	50.3%	46.2%	43.1%	**48.3%**

3. Additional analysis

and non-parametric evidence

Figure S2 shows a locally weighted (LOWESS) estimate of the relationship between RiskD and age-residualized cognitive ability across the full sample. The relationship is markedly non-linear: RiskD is broadly flat below the sample mean ($CogAb(r) = 0$) and rises above it. This motivates the quadratic term $CogAb(r)^2$ in columns 4 and 8 of Table 1, which is positive and significant for RiskD but small and insignificant for PayD. By contrast, we find no evidence that the age gradient in RiskD differs materially by cognitive ability: the interaction between age and $CogAb(r)$ is not statistically significant, whether ability enters continuously ($F_{1,2089} = 1.74$, $p = 0.19$) or as a median split within each age group ($F_{1,2089} = 0.30$, $p = 0.59$), and the implied age slopes are similar (0.022 among low-ability and 0.026 among high-ability participants). Cognitive ability is therefore strongly associated with the level of risk-based coordination, and non-linearly so, while these specifications provide no evidence that it materially alters the age profile.

Figure S3 reports the age-specific gender difference in RiskD implied by the linear probability model including the $Age \times Female$ interaction. The point estimates become more negative with age, but the underlying interaction is significant only at the 10% level and does not survive the logit specification (Table S21). We therefore present the figure as descriptive evidence on the gender difference in levels and do not interpret its age pattern as evidence of distinct age trajectories by gender.

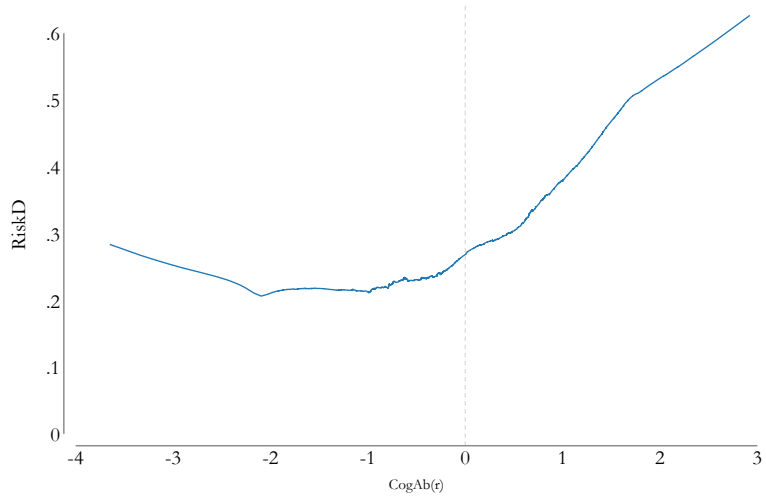


Fig. S2. Locally weighted (LOWESS) relationship between risk-based coordination and cognitive ability. Smoothed regression of RiskD on age-residualized cognitive ability across the full sample (bandwidth $= 0.6$). The dashed vertical line marks the sample mean, $CogAb(r) = 0$. The upper tail of the ability distribution is sparsely populated: 1% of participants have $CogAb(r) > 2$.

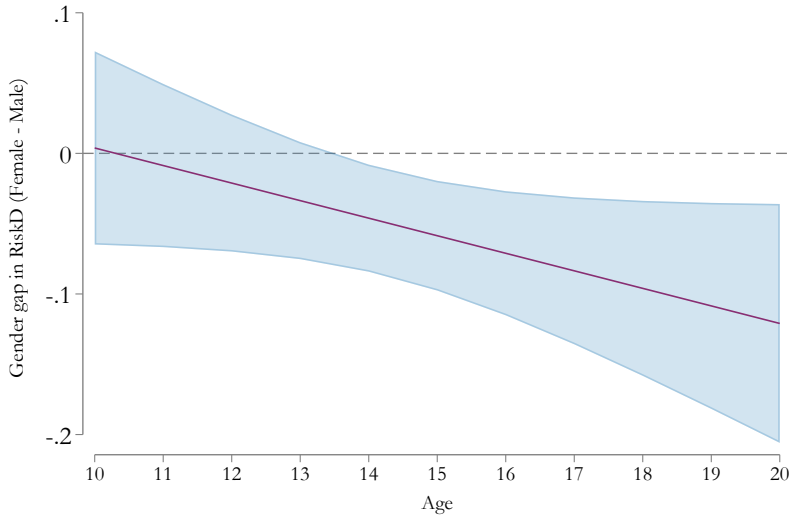


Fig. S3. Gender gap in risk-based coordination, by age. Average marginal effect of *Female* on RiskD at each age, with 95% confidence intervals, estimated from a linear probability model including the $Age \times Female$ interaction, age-residualized cognitive ability, school fixed effects, and baseline controls (repeater status, class size, incentive condition), with robust standard errors. Negative values indicate lower rates of RiskD among females. Because the marginal effect is linear in age by construction, the profile should not be interpreted as identifying a change point or a distinct age trajectory. The underlying $Age \times Female$ interaction is not robust across specifications.

A. The difference in age gradients across games. The central claim of the paper is comparative: the age gradient differs between the two games. This section reports the formal test of that difference and examines the most direct alternative account, namely that the stronger age gradient in the risk-dominance game reflects age-related differences in task difficulty or confusion.

Panel A of Table S9 pools the two games, giving two observations per participant: the dependent variable is PayD for the Game 1 observation and RiskD for the Game 2 observation. All covariates are interacted with a Game 2 indicator, and standard errors are clustered at the participant level. Because age is not centered, the main effect of Game 2 is evaluated at age zero, outside the observed age range, and we do not interpret it substantively. The coefficient of primary interest is the Age × Game 2 interaction, which is positive and significant (0.011, $p = 0.011$), showing directly that the age gradient is steeper in Game 2. By contrast, the CogAb(r) × Game 2 interaction is small and statistically insignificant (0.004, $p = 0.77$), so we do not detect a difference between the games in the within-age cognitive-ability gradient. Thus, while the two tasks differ in several respects, the stronger age gradient in Game 2 is not mirrored by a stronger gradient in the cognitive-ability measure available in our data.

A second possibility is that younger participants are relatively more confused by Game 2 and that this difference in confusion declines with age. Panel B of Table S9 examines this possibility using a proxy for non-strategic engagement defined without

reference to the chosen action. The indicator equals one when the participant reports "I liked the color" or "I chose one randomly". These reasons are considerably more frequent in Game 2 than in Game 1 (44.5% vs. 30.4%), consistent with greater difficulty in the risk-dominance game. Column 1 regresses the within-participant difference between the two games on age and the baseline controls. We find no evidence that this gap declines with age (-0.004, $p = 0.73$). Column 2 repeats the exercise with a broader proxy that also counts internally inconsistent action–reason pairs—pairs in which the stated reason contradicts the action taken, defined in Section C—and likewise provides no evidence of a declining gap (-0.015, $p = 0.21$). We report this broader measure only as a robustness check because, in Game 2, its complement partly overlaps with RiskD itself. Taken together, these proxies do not display the age pattern that would be expected if declining relative confusion in Game 2 alone generated the RiskD gradient.

Table S9. Age gradients and proxies for relative task difficulty across games.

Panel A. Stacked model

	(1) PayD/RiskD
Age	0.016**
	(0.007)
CogAb(r)	0.068***
	(0.010)
Female	−0.047**
	(0.021)
Game 2	−0.479***
	(0.067)
Age × Game 2	0.011**
	(0.004)
CogAb(r) × Game 2	0.004
	(0.013)
Female × Game 2	−0.006
	(0.026)
Constant	0.595***
	(0.117)
Observations	4,249
R-squared	0.158
School FE	Y
Controls	Y

Panel B. Difficulty gap, Game 2 *vs* Game 1

	(1) Non-strategic	(2) + inconsistent
Age	−0.004	−0.015
	(0.012)	(0.012)
CogAb(r)	−0.010	−0.002
	(0.013)	(0.013)
Female	0.021	0.029
	(0.026)	(0.027)
Constant	0.235	0.576***
	(0.185)	(0.189)
Observations	2,110	2,110
R-squared	0.024	0.023
School FE	Y	Y
Controls	Y	Y

Note: **Panel A** stacks the two games, giving two observations per participant: the dependent variable is PayD for the Game 1 observation and RiskD for the Game 2 observation. All covariates are interacted with a Game 2 indicator, and standard errors are clustered at the participant level. Because age is not centered, the Game 2 main effect is evaluated at age zero and is not substantively interpreted. **Panel B** uses as dependent variable the within-participant difference between Game 2 and Game 1 in an indicator of *non-strategic justification*, which equals one when the participant reports "I liked the color" or "I chose one randomly". This indicator is defined without reference to the chosen action, although it is constructed from the same stated-reason item as the strict measures. Column 2 uses a broader version of the same indicator that also counts internally inconsistent action–reason pairs, that is, pairs in which the stated reason contradicts the action taken; we report it only as a robustness check because, in Game 2, its complement partly overlaps with RiskD itself. Panel B is restricted to participants who recorded a justification in both games and uses robust standard errors. All specifications include school fixed effects and baseline controls (repeater status, class size, incentive condition). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B. Coordination types. The two criteria may combine in different ways across participants. To examine their joint distribution, we classify each participant into one of three mutually exclusive and exhaustive types: satisfying *neither* criterion (PayD = 0 and RiskD = 0), *exactly one* of them (PayD = 1 or RiskD = 1, but not both), or *both* (PayD = 1 and RiskD = 1). Across the sample, 32.5% satisfy neither, 45.2% exactly one, and 22.3% both.

Participants who satisfy exactly one criterion almost always satisfy PayD rather than RiskD: 85.1% of that category has PayD = 1 and RiskD = 0. In the full sample, 38.4% satisfy PayD but not RiskD, compared with 6.8% who satisfy RiskD but not PayD. Satisfying the risk-based criterion without the payoff-based criterion is therefore relatively uncommon.

Figure S4 shows how the three categories vary with age. Panel A uses the strict measures. Panels B and C repeat the classification using each component separately: participants' actions in Panel B and their stated reasons in Panel C. In all three panels, the main age-related movement occurs at the extremes: the share satisfying both criteria increases, the share satisfying neither decreases, and the intermediate category changes comparatively little.

Table S10 quantifies these profiles. Because the three indicators are exhaustive and mutually exclusive, the coefficients on any covariate sum to zero across columns. Using the strict measures (Panel A), the share satisfying neither criterion declines with age (-0.025, $p = 0.006$), the share satisfying both increases (0.018, $p = 0.029$), and the intermediate category shows no statistically significant age gradient (0.008, $p = 0.44$). The increase in the "both" category is similar in magnitude when the classification is based on actions alone (0.023, $p = 0.019$) or on stated reasons alone (0.017, $p = 0.069$). The decline in the "neither" category is weaker in the stated-reason decomposition (-0.011, $p = 0.19$) than in the other two. Cognitive ability is strongly associated with a shift from the "neither" to the "both" category. Females are less likely to fall in the "both" category

$(-0.072,\ p < 0.01)$, with corresponding positive coefficients for the other two categories.

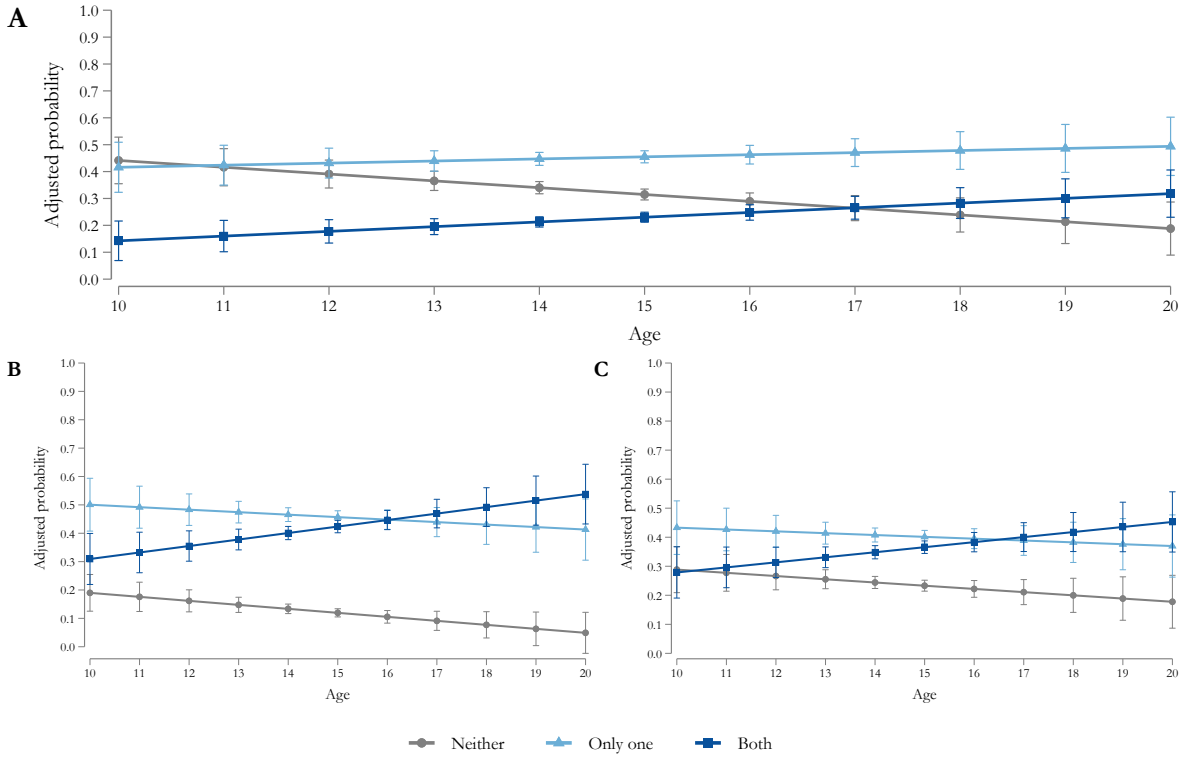


Fig. S4. Age profiles of coordination criteria satisfied. Adjusted predicted probabilities of satisfying neither, exactly one, or both criteria, with 95% confidence intervals, from linear probability models controlling for age-residualized cognitive ability, gender, school fixed effects, and baseline controls (repeater status, class size, incentive condition). Age enters linearly, so the profiles are linear by construction. **(A)** Classification by the strict measures PayD and RiskD ($N = 2{,}118$). **(B)** Classification by actions alone, P and R ($N = 2{,}118$). **(C)** Classification by stated reasons alone, "I earned more" and "It was less risky" ($N = 2{,}110$).

Table S10. Determinants of coordination types.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	A) Strict measures (PayD, RiskD)			B) Actions only (P, R)			C) Stated reasons only		
	Neither	One	Both	Neither	One	Both	Neither	One	Both
Age	−0.025***	0.008	0.018**	−0.014**	−0.009	0.023**	−0.011	−0.006	0.017*
	(0.009)	(0.010)	(0.008)	(0.007)	(0.010)	(0.010)	(0.008)	(0.010)	(0.010)
CogAb(r)	−0.068***	−0.003	0.071***	−0.030***	−0.032***	0.062***	−0.045***	−0.024**	0.069***
	(0.010)	(0.011)	(0.009)	(0.007)	(0.011)	(0.010)	(0.009)	(0.011)	(0.010)
Female	0.030	0.042*	−0.072***	0.018	0.041*	−0.058***	0.031	0.044**	−0.074***
	(0.020)	(0.022)	(0.018)	(0.014)	(0.022)	(0.021)	(0.019)	(0.022)	(0.021)
Constant	0.486***	0.318**	0.196	0.266**	0.453***	0.282*	0.253*	0.405***	0.342**
	(0.144)	(0.156)	(0.127)	(0.103)	(0.155)	(0.151)	(0.132)	(0.154)	(0.149)
Observations	2,118	2,118	2,118	2,118	2,118	2,118	2,110	2,110	2,110
R-squared	0.077	0.032	0.083	0.044	0.030	0.064	0.040	0.016	0.061
School FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: Linear probability models. Each block of three columns classifies participants into three mutually exclusive and exhaustive types according to a pair of binary indicators: *Neither* means both indicators equal zero, *One* means exactly one of them equals one, and *Both* means both equal one. Because the three types are exhaustive and mutually exclusive, the coefficients on any covariate sum to zero across each block. Columns 1–3 use the strict measures PayD and RiskD; columns 4–6 use the actions alone, P and R; columns 7–9 use the stated reasons alone, "I earned more" in Game 1 and "It was less risky" in Game 2. Columns 7–9 have fewer observations because eight participants recorded an action in Game 2 without a justification. All specifications include school fixed effects and baseline controls (repeater status, class size, incentive condition). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C. Validity of the strict measures. PayD and RiskD each pair an action with a stated reason chosen from a closed list of four options after the action has been recorded. This section reports what that stated-reason component captures. We use three constructed variables. An *internally inconsistent pair* is one in which the stated reason contradicts the chosen action: in Game 1, the payoff-dominant action with "I earned less", or the alternative action with "I earned more"; in Game 2, the risk-dominant action with "It was more risky", or the alternative action with "It was less risky". *Misuse of the risk vocabulary* decomposes the Game 2 inconsistencies by the action they follow, so that each rate is conditional on a given choice. Finally, we decompose each strict measure into an *action margin* and a *conditional reason margin*, following the order of the experiment.

Stated reasons are generally consistent with actions. Strictly contradictory pairs occur in 7.5% of responses in Game 1 and 21.0% in Game 2, consistent with greater difficulty in the risk-dominance game. We find no evidence that inconsistency declines with age in either game (Table S11). In Game 1, inconsistency is negatively associated with cognitive ability instead (-0.020, $p < 0.01$), suggesting that task processing may contribute to these inconsistencies rather than age-related differences in vocabulary alone.

Table S11. Inconsistent action–reason pairs show no detectable decline with age.

	(1)	(2)
	Inconsistent pair	
	Game 1	Game 2
Age	0.000	−0.011
	(0.005)	(0.008)
CogAb(r)	−0.020***	−0.011
	(0.006)	(0.009)
Female	−0.005	0.003
	(0.011)	(0.018)
Constant	0.038	0.389***
	(0.082)	(0.130)
Observations	2,131	2,110
R-squared	0.046	0.025
School FE	Y	Y
Controls	Y	Y

Note: The dependent variable equals one if the stated reason contradicts the chosen action. In Game 1, inconsistent pairs are the payoff-dominant action with "I earned less" or the alternative action with "I earned more". In Game 2, they are the risk-dominant action with "It was more risky" or the alternative action with "It was less risky". All specifications include school fixed effects and baseline controls (repeater status, class size, incentive condition). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The responses also show cross-game consistency. The within-participant correlation between PayD and RiskD is 0.21 ($p < 0.001$). Participants reporting a random choice in both games (6.9%) or a color-based choice in both games (6.1%) are about twice as common as they would be under independent selection across the two screens (3.3% and 3.4%). Figure S5 shows the full composition of stated reasons by action and age. Among participants choosing the relevant action, the congruent reason is the modal response at every age, including the youngest: among ten-year-olds choosing the risk-dominant action, 58.8% select "It was less risky". This does not establish causality or the process generating the choice, but it shows that stated reasons covary with actions in a systematic way across the age range.

A more specific concern is that RiskD tracks familiarity with the word "risk". If age-related changes in comprehension of that vocabulary were driving the RiskD gradient, we would expect incorrect uses of the term to become less frequent with age. We find no evidence of such a decline (Table S12). Among participants who chose the non-risk-dominant action, 32.7% nonetheless report "It was less risky", with no detectable age gradient (-0.012, $p = 0.40$); the mirror error—choosing the risk-dominant action and reporting "It was more risky"—occurs in 10.6% of cases and likewise shows no detectable age gradient (-0.001, $p = 0.91$). These results provide little support for an account in which growing familiarity with the risk vocabulary generates the observed RiskD age gradient.

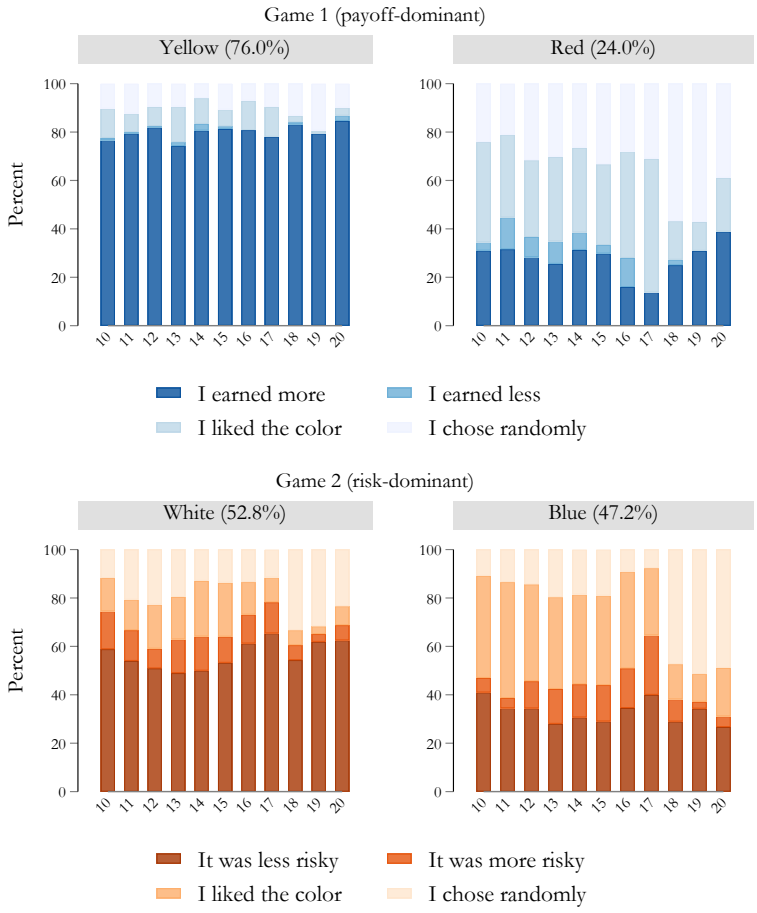


Fig. S5. Composition of stated reasons by chosen action and age. Each bar shows the distribution of stated reasons among participants who chose the indicated action, by age. The congruent reason is "I earned more" in Game 1 and "It was less risky" in Game 2; the opposite reason is "I earned less" and "It was more risky", respectively. Panel headers report the overall share of participants choosing each action. $N = 2{,}131$ in Game 1 and $N = 2{,}110$ in Game 2. Among participants choosing the risk-dominant action, the congruent reason is modal from age 10 onwards. Cell sizes for the non-selected actions are small at some ages, in particular ages 16–17 for Red.

Table S12. No detectable age decline in misuse of the risk vocabulary.

	(1) Reported "less risky" among those choosing Blue	(2) Reported "more risky" among those choosing White
Age	−0.012	−0.001
	(0.014)	(0.009)
CogAb(r)	0.016	−0.017*
	(0.016)	(0.009)
Female	−0.015	0.017
	(0.030)	(0.019)
Constant	0.586***	0.146
	(0.221)	(0.138)
Observations	994	1,116
R-squared	0.042	0.033
School FE	Y	Y
Controls	Y	Y

Note: Column 1 restricts the sample to participants who chose Blue, the non-risk-dominant action in Game 2; the dependent variable equals one if they reported "It was less risky". Column 2 restricts the sample to participants who chose White, the risk-dominant action; the dependent variable equals one if they reported "It was more risky". If age-related changes in comprehension of the risk vocabulary were driving the RiskD gradient, both misuse rates would be expected to decline with age; we find no evidence that either does. All specifications include school fixed effects and baseline controls (repeater status, class size, incentive condition). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Finally, the central age pattern does not rest on the verbal report alone. Because each strict measure is a conjunction, its probability can be written descriptively as the product of an action margin and a conditional reason margin, following the order of the experiment. Table S13 estimates both. The age gradient is present in the action margin, which involves no language: the probability of choosing the risk-dominant action rises with age (0.027, $p = 0.007$; column 4). The conditional reason margin is positive but imprecisely estimated (0.021, $p = 0.13$; column 5). In the payoff game, neither margin shows a statistically significant age gradient (columns 1 and 2). Columns 2 and 5 condition on having taken a particular action and are therefore estimated on endogenously selected subsamples, so the decomposition is descriptive rather than causal. This does not remove the limitations of a stated-reason measure, but it shows that the RiskD age pattern is also visible in the action itself.

Table S13. Margin decomposition: the age gradient is present in the action margin.

	(1)	(2)	(3)	(4)	(5)	(6)
	(A) Game 1 (payoff-dominant)			**(B)** Game 2 (risk-dominant)		
	P	"Earned more" | P = 1	PayD	R	"Less risky" | R = 1	RiskD
Age	0.010	0.014	0.019*	0.027***	0.021	0.024***
	(0.009)	(0.009)	(0.010)	(0.010)	(0.014)	(0.009)
CogAb(r)	0.046***	0.039***	0.068***	0.044***	0.086***	0.072***
	(0.009)	(0.010)	(0.010)	(0.011)	(0.014)	(0.010)
Female	−0.054***	−0.006	−0.046**	−0.018	−0.084***	−0.053***
	(0.018)	(0.020)	(0.021)	(0.022)	(0.030)	(0.019)
Constant	0.710***	0.796***	0.581***	0.301*	0.389*	0.128
	(0.132)	(0.144)	(0.149)	(0.154)	(0.213)	(0.139)
Observations	2,131	1,620	2,131	2,118	1,119	2,118
R-squared	0.068	0.049	0.078	0.047	0.085	0.065
Sample	Full	P = 1	Full	Full	R = 1	Full
School FE	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y

Note: P denotes the payoff-dominant action in Game 1 and R the risk-dominant action in Game 2; R is not a color label. Columns 1 and 4 use the full sample and indicate whether the participant chose P or R. Columns 2 and 5 restrict the sample to participants who chose that action and indicate whether they selected the congruent reason; participants who chose the relevant action without recording a justification are coded as not having given the congruent reason, consistent with the construction of PayD and RiskD. Columns 3 and 6 report the joint measures PayD and RiskD. Because columns 2 and 5 are estimated on endogenously selected subsamples, the decomposition is descriptive. All specifications include school fixed effects and baseline controls (repeater status, class size, incentive condition). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D. Beliefs about the partner. Risk dominance is behaviorally meaningful only under uncertainty about what the partner will do, so a natural check on the strict measures is whether they are associated with participants' beliefs. In the six schools described in Section 1B, participants also reported which button they expected their partner to press and which reason they attributed to the partner. This lets us pair each measure with its belief counterpart: P and R with beliefs about the partner's action, and PayD and RiskD with the conjunction of the expected action and the attributed reason.

Table S14 reports the results. Behavior is strongly associated with beliefs. Participants who expect the partner to take the relevant action are substantially more likely to take it themselves (0.327 in Game 1 and 0.400 in Game 2, both $p < 0.001$), and those who both expect that action and attribute the congruent reason to the partner are substantially more likely to satisfy the strict measure themselves (0.577 for PayD and 0.475 for RiskD, both $p < 0.001$).

The randomized order of the own-choice and belief blocks also speaks to the concern that stated reasons are constructed after the fact. A belief reported before the decision cannot rationalize a choice not yet made. We do not detect a statistically significant Belief × Belief-first interaction in any column, indicating that the belief–behavior association is of similar magnitude under the two elicitation orders.

Whether elicitation order shifted the level of each variable is a separate question, and one the specification in Table S14 does not answer directly. We therefore regress each variable on the belief-first indicator alone, keeping the same controls but dropping the belief and its interaction. In Game 1, participants who reported their beliefs first were more likely to satisfy PayD (+0.125, $p = 0.008$) and more likely to expect the partner to press the payoff-dominant button (+0.155, $p = 0.001$). We do not detect comparable order effects in Game 2. This pattern is consistent with priming: asking participants to reason about their partner may make the strategic dimension of the task more salient. It also motivates the action-first design of the main experiment, in which the action was recorded before the list of reasons was displayed.

This subsample is small, comes from six schools, and covers only part of the age range, so we do not use it to characterize how beliefs vary with age. Moreover, belief–choice consistency may partly reflect projection: participants who choose an action may expect their partner to do the same. We therefore interpret these results as evidence of consistency between beliefs and behavior, not as causal evidence that beliefs determine choices.

E. Sample composition and the age gradient across studies. Our sample combines a classroom study with a laboratory study of older participants, so age is partly confounded with study context: older participants are not simply older versions of the

Table S14. Coordination is associated with beliefs about the partner (belief subsample).

	(1)	(2)	(3)	(4)
	Game 1 (payoff-dominant)		Game 2 (risk-dominant)	
	P	PayD	R	RiskD
Belief about partner	0.327***	0.577***	0.400***	0.475***
	(0.069)	(0.061)	(0.066)	(0.073)
Belief elicited first	−0.075	0.049	0.086	−0.052
	(0.080)	(0.053)	(0.071)	(0.041)
Belief × Belief first	0.085	0.021	−0.124	0.148
	(0.095)	(0.075)	(0.096)	(0.098)
Constant	0.068	−0.040	0.624	0.652*
	(0.403)	(0.339)	(0.425)	(0.335)
Observations	403	403	394	394
R-squared	0.242	0.462	0.166	0.359
School FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Note: Belief-elicitation subsample (six schools of Study 1; ages 12–15 and 18–20). Each dependent variable is paired with its counterpart elicited about the partner, as reported by the participant; the partner's actual behavior is not used. In columns 1 and 3 the belief variable indicates that the participant expected the partner to press the equilibrium button (Yellow in Game 1, White in Game 2); in columns 2 and 4 it indicates that the participant both expected that action and attributed the congruent reason to the partner. "Belief elicited first" indicates that the partner-belief block preceded the participant's own decision; the order was randomized at the individual level (belief-first arm: $N = 213$ in Game 1 and $N = 197$ in Game 2). The belief coefficient therefore corresponds to the arm in which beliefs were elicited after the own decision, and the interaction measures the change in the belief-first arm; we do not detect a statistically significant interaction in any column. Given the restricted age support and sample size, we do not use these data to characterize age differences in beliefs. All specifications include age, age-residualized cognitive ability, gender, school fixed effects and baseline controls (repeater status, class size, incentive condition). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

younger participants. This section examines whether the RiskD age gradient is also present within Study 1 and whether it differs detectably across studies.

Within Study 1 alone, which uses the same platform and instructions throughout and spans ages 10 to 20 through its vocational-training and university-pilot participants, the age coefficient for RiskD is 0.021 ($p = 0.020$; Table S15, Panel A). The corresponding coefficient for the risk-dominant action alone is 0.031 ($p = 0.003$). By contrast, we do not detect a statistically significant age gradient for PayD (0.012, $p = 0.22$) or for the payoff-dominant action (0.009, $p = 0.32$). Thus, the positive RiskD age gradient is present within Study 1 alone.

Panel B examines whether the age gradient differs between studies by interacting age with a Study 2 indicator in the pooled sample. We do not detect a statistically significant interaction for either RiskD (0.050, $p = 0.20$) or PayD (0.040, $p = 0.29$), while the RiskD age coefficient remains positive and significant (0.022, $p = 0.015$). The Study 2 main effect is nearly collinear with the school fixed effects and is estimated imprecisely, so we focus on the interaction. These estimates provide no evidence that the RiskD age gradient is generated by a different age slope in Study 2.

Table S15. Age gradients within Study 1 and across studies.

	(1)	(2)	(3)	(4)
	(A) Study 1 only		**(B)** Pooled, Age × Study 2	
	PayD	RiskD	PayD	RiskD
Age	0.012	0.021**	0.017*	0.022**
	(0.010)	(0.009)	(0.010)	(0.009)
Study 2			−0.804	−1.075
			(0.725)	(0.752)
Age × Study 2			0.040	0.050
			(0.038)	(0.039)
CogAb(r)	0.088***	0.084***	0.068***	0.072***
	(0.013)	(0.012)	(0.010)	(0.010)
Female	−0.043*	−0.025	−0.045**	−0.051***
	(0.023)	(0.021)	(0.021)	(0.019)
Constant	0.474***	0.136	0.618***	0.171
	(0.162)	(0.154)	(0.155)	(0.143)
Observations	1,716	1,703	2,131	2,118
R-squared	0.092	0.068	0.079	0.067
School FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Note: **Panel A** restricts the sample to Study 1, which uses a single protocol and spans ages 10 to 20 through its vocational-training and university-pilot participants. **Panel B** estimates the pooled model interacting age with a Study 2 indicator. The Study 2 main effect is nearly collinear with the school fixed effects and is therefore estimated imprecisely; the coefficient of interest is the interaction. The corresponding pooled baseline estimates are reported in Table 1: 0.019 for PayD and 0.024 for RiskD. All specifications include age-residualized cognitive ability, gender, school fixed effects and baseline controls (repeater status, class size, incentive condition). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Finally, Tables S16 to S18 re-estimate the main specifications on progressively broader school-based samples: late primary and secondary; adding high school; and adding vocational training. The RiskD age coefficient is 0.015, 0.021, and 0.021 across these samples, compared with 0.024 in the full sample. The point estimates remain within a relatively narrow range as age support and precision increase. PayD shows no statistically significant age gradient in any of these restricted samples. Taken together with the Study 1 estimates above, these results provide little indication that combining the two studies is responsible for the positive RiskD age gradient.

Table S16. OLS regression for the sub-sample: Late primary + Secondary.

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.009	0.009	0.007	0.009	0.015	0.015	0.015	0.015
	(0.011)	(0.011)	(0.013)	(0.011)	(0.010)	(0.010)	(0.012)	(0.010)
CogAb(r)	0.093***	−0.078	0.093***	0.092***	0.087***	0.014	0.087***	0.087***
	(0.014)	(0.119)	(0.014)	(0.014)	(0.013)	(0.103)	(0.013)	(0.013)
Female	−0.035	−0.034	−0.092	−0.037	−0.021	−0.020	−0.008	−0.020
	(0.025)	(0.025)	(0.192)	(0.025)	(0.023)	(0.023)	(0.173)	(0.023)
Age × CogAb(r)		0.013				0.006		
		(0.009)				(0.008)		
Age × Female			0.004				−0.001	
			(0.015)				(0.013)	
CogAb(r)^2				−0.014				0.005
				(0.013)				(0.011)
Constant	0.425*	0.411*	0.451*	0.427*	0.179	0.174	0.174	0.179
	(0.227)	(0.227)	(0.244)	(0.226)	(0.208)	(0.209)	(0.221)	(0.208)
Observations	1,474	1,474	1,474	1,474	1,462	1,462	1,462	1,462
R-squared	0.083	0.085	0.083	0.084	0.062	0.063	0.062	0.063
BIC	2108.8	2113.9	2116.0	2114.9	1812.6	1819.4	1819.9	1819.7
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Note: In columns 2 and 6 the CogAb(r) coefficient is evaluated at age zero and is therefore not comparable to the other columns. All specifications include school fixed effects and baseline controls (repeater status, class size, and incentive condition). Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table S17. OLS regression for the sub-sample: Late primary + Secondary + High School.

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.015	0.015	0.012	0.015	0.021**	0.021**	0.029***	0.021**
	(0.010)	(0.010)	(0.012)	(0.010)	(0.009)	(0.009)	(0.011)	(0.009)
CogAb(r)	0.085***	0.073	0.085***	0.084***	0.080***	0.122	0.080***	0.081***
	(0.013)	(0.095)	(0.013)	(0.013)	(0.012)	(0.085)	(0.012)	(0.012)
Female	−0.032	−0.031	−0.117	−0.033	−0.040*	−0.040*	0.168	−0.038*
	(0.023)	(0.024)	(0.161)	(0.024)	(0.022)	(0.022)	(0.147)	(0.022)
Age × CogAb(r)		0.001				−0.003		
		(0.007)				(0.006)		
Age × Female			0.006				−0.015	
			(0.012)				(0.011)	
CogAb(r)^2				−0.009				0.011
				(0.011)				(0.010)
Constant	0.588***	0.587***	0.629***	0.594***	0.114	0.118	0.013	0.106
	(0.161)	(0.161)	(0.179)	(0.161)	(0.149)	(0.150)	(0.165)	(0.149)
Observations	1,666	1,666	1,666	1,666	1,653	1,653	1,653	1,653
R-squared	0.076	0.076	0.076	0.077	0.060	0.060	0.061	0.061
BIC	2397.8	2405.2	2404.9	2404.5	2097.3	2104.4	2102.6	2103.4
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Note: In columns 2 and 6 the CogAb(r) coefficient is evaluated at age zero and is therefore not comparable to the other columns. All specifications include school fixed effects and baseline controls (repeater status, class size, and incentive condition). Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table S18. OLS regression for the sub-sample: Late primary + Secondary + High School + Vocational.

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.016	0.016	0.016	0.016	0.021**	0.021**	0.024**	0.021**
	(0.010)	(0.010)	(0.011)	(0.010)	(0.009)	(0.009)	(0.010)	(0.009)
CogAb(r)	0.084***	0.087	0.084***	0.083***	0.082***	0.100	0.081***	0.083***
	(0.013)	(0.082)	(0.013)	(0.013)	(0.012)	(0.073)	(0.012)	(0.012)
Female	−0.036	−0.036	−0.043	−0.038*	−0.034	−0.034	0.042	−0.032
	(0.023)	(0.023)	(0.140)	(0.023)	(0.021)	(0.021)	(0.124)	(0.021)
Age × CogAb(r)		−0.000				−0.001		
		(0.006)				(0.005)		
Age × Female			0.001				−0.006	
			(0.010)				(0.009)	
CogAb(r)^2				−0.013				0.014
				(0.011)				(0.010)
Constant	0.572***	0.572***	0.575***	0.581***	0.108	0.109	0.070	0.097
	(0.159)	(0.159)	(0.173)	(0.159)	(0.147)	(0.148)	(0.159)	(0.148)
Observations	1,777	1,777	1,777	1,777	1,764	1,764	1,764	1,764
R-squared	0.084	0.084	0.084	0.085	0.061	0.061	0.062	0.063
BIC	2561.2	2568.7	2568.7	2567.3	2233.1	2240.5	2240.2	2238.3
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Note: In columns 2 and 6 the CogAb(r) coefficient is evaluated at age zero and is therefore not comparable to the other columns. All specifications include school fixed effects and baseline controls (repeater status, class size, and incentive condition). Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

F. Flexible age specifications. Our main regressions treat age linearly, which for a developmental paper should be assessed rather than assumed. We report three checks on that specification.

First, adding a full set of age indicators to the linear specification does not significantly improve fit for either outcome (PayD: $F_{9,2094} = 1.46$, $p = 0.16$; RiskD: $F_{9,2081} = 0.68$, $p = 0.73$). With school fixed effects included, individual age indicators above 17 are nearly collinear with the Study 2 school indicators, so we rely on the joint test rather than on individual coefficients.

Second, we estimate a linear spline with a knot at age 16, the end of compulsory secondary education in Spain (Table S19).

For RiskD, the slope from ages 10 to 16 is positive and significant (0.022, $p = 0.018$), while the post-16 slope is larger in point estimate but imprecisely estimated (0.040, $p = 0.15$). Equality of the two slopes is not rejected ($p = 0.54$). For PayD, neither slope is statistically significant and equality is likewise not rejected ($p = 0.41$). We therefore find no evidence that the RiskD age gradient is concentrated specifically after age 16.

Third, adding a quadratic term in age yields insignificant quadratic terms for both outcomes. Taken together, these checks provide no evidence against using a linear age term as a parsimonious summary of the age profiles over the observed range. This is distinct from the non-linear association with cognitive ability documented above: in the baseline data the quadratic term in $CogAb(r)$ is significant for RiskD, whereas the quadratic term in age is not.

One comparison deserves clarification. The spline gives a pre-16 RiskD slope of 0.022 ($p = 0.018$), whereas the late-primary-plus-secondary subsample in Table S16, which covers approximately the same ages, gives 0.015 ($p = 0.13$). The samples are not identical: the latter is defined by educational stage, whereas the spline uses the full sample to estimate fixed effects and controls. The two point estimates are close relative to their sampling uncertainty.

Table S19. Flexible age specifications.

	(1)	(2)	(3)	(4)
	(A) Linear spline		**(B)** Quadratic	
	PayD	RiskD	PayD	RiskD
Age slope, 10–16	0.016	0.022**		
	(0.010)	(0.009)		
Age slope, 17–20	0.040	0.040		
	(0.027)	(0.028)		
Age			−0.061	−0.013
			(0.063)	(0.061)
Age squared			0.003	0.001
			(0.002)	(0.002)
CogAb(r)	0.067***	0.072***	0.068***	0.072***
	(0.010)	(0.010)	(0.010)	(0.010)
Female	−0.046**	−0.053***	−0.046**	−0.053***
	(0.021)	(0.019)	(0.021)	(0.019)
Constant	0.623***	0.160	1.131**	0.379
	(0.159)	(0.145)	(0.449)	(0.434)
Observations	2,131	2,118	2,131	2,118
R-squared	0.079	0.065	0.079	0.065
School FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Note: **Panel A** estimates a linear spline in age with a knot at 16, the end of compulsory secondary education in Spain; the two coefficients are the slopes on either side of the knot, not a level shift, and the fitted profile is continuous at 16. Wald tests do not reject equality of the two slopes (PayD: $p = 0.41$; RiskD: $p = 0.54$). **Panel B** includes age and age squared. A nested test comparing age indicators with the linear specification likewise provides no evidence against the linear age specification (PayD: $F_{9,2094} = 1.46$, $p = 0.16$; RiskD: $F_{9,2081} = 0.68$, $p = 0.73$). Because school fixed effects are included, age indicators above 17 are nearly collinear with the Study 2 school indicators, so we report the joint test rather than the individual coefficients. All specifications include age-residualized cognitive ability, gender, school fixed effects and baseline controls (repeater status, class size, incentive condition). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

G. Further robustness. This section reports three further checks that yield qualitatively similar main estimates: a triple interaction between age, gender, and cognitive ability; logit specifications reported as average marginal effects; and a replication of Table 1 that replaces the CRT rescaling for Study 2 participants with multiple imputation.

Table S20 adds the triple interaction between age, gender, and cognitive ability, together with all three two-way interactions. The triple interaction is small and not statistically significant for either outcome (0.011, $p = 0.13$ for PayD; 0.005, $p = 0.45$ for RiskD), providing no evidence that the age-by-gender pattern differs systematically with cognitive ability. Table S21 reports logit specifications as average marginal effects; the resulting estimates are qualitatively similar to those from the linear probability models.

Table S22 replicates Table 1 using MICE-PMM imputation ($M = 20$). The main pattern is qualitatively unchanged: the RiskD age coefficient remains positive and statistically significant across specifications, while the PayD age gradient remains weaker. The estimated associations with cognitive ability are also similar in the directly comparable specifications. Overall, the alternative treatment of the cognitive-ability measure does not materially alter the principal results.

Table S20. OLS regressions of strict coordination with a triple interaction between age, gender and cognitive ability.

	(1) PayD	(2) PayD	(3) RiskD	(4) RiskD
Age	0.021**	0.020**	0.030***	0.029***
	(0.010)	(0.010)	(0.009)	(0.009)
CogAb(r)	0.237***	0.232***	0.160**	0.128*
	(0.070)	(0.072)	(0.067)	(0.067)
CogAb(r)^2		0.003		0.016**
		(0.008)		(0.008)
Female	0.030	0.029	0.127	0.120
	(0.108)	(0.108)	(0.099)	(0.099)
Age × CogAb(r)	−0.010**	−0.010**	−0.006	−0.004
	(0.004)	(0.004)	(0.004)	(0.004)
Age × Female	−0.005	−0.005	−0.012*	−0.012*
	(0.007)	(0.007)	(0.007)	(0.007)
Female × CogAb(r)	−0.182	−0.181	−0.061	−0.051
	(0.112)	(0.112)	(0.103)	(0.103)
Age × Female × CogAb(r)	0.011	0.011	0.005	0.004
	(0.007)	(0.007)	(0.007)	(0.007)
Constant	0.553***	0.551***	0.048	0.037
	(0.158)	(0.158)	(0.147)	(0.147)
Observations	2,131	2,131	2,118	2,118
R-squared	0.081	0.081	0.068	0.070
BIC	3058.4	3065.9	2763.3	2765.9
School FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Note: All specifications include the three two-way interactions among age, gender and cognitive ability alongside the triple interaction. Because age is not centered, the coefficients on CogAb(r), Female and their interaction are evaluated at age zero and are not effects at any observed age; the coefficient of interest is the triple interaction. Columns 2 and 4 add a quadratic term in CogAb(r). All specifications include school fixed effects and baseline controls (repeater status, class size, incentive condition). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table S21. Logit regressions of strict coordination on age, cognitive ability, and gender.

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.019**	0.018*	0.021**	0.018**	0.025***	0.025***	0.029***	0.025***
	(0.009)	(0.009)	(0.010)	(0.009)	(0.009)	(0.009)	(0.010)	(0.009)
CogAb(r)	0.067***	0.156***	0.067***	0.071***	0.069***	0.170***	0.068***	0.072***
	(0.010)	(0.055)	(0.010)	(0.011)	(0.010)	(0.052)	(0.010)	(0.009)
Female	−0.048**	−0.049**	0.038	−0.046**	−0.051***	−0.052***	0.108	−0.047**
	(0.021)	(0.021)	(0.108)	(0.021)	(0.019)	(0.019)	(0.102)	(0.019)
Age × CogAb(r)		−0.006*				−0.006*		
		(0.003)				(0.003)		
Age × Female			−0.006				−0.011	
			(0.007)				(0.007)	
CogAb(r)^2				0.009				0.016**
				(0.008)				(0.007)
Observations	2,131	2,131	2,131	2,131	2,118	2,118	2,118	2,118
CogAbB2	No	No	No	Yes	No	No	No	Yes
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Note: Average marginal effects from logit models, computed as the average over the estimation sample of the marginal effect of each variable. In columns 2 and 6 the CogAb(r) marginal effect is evaluated at age zero and is therefore not comparable to the other columns. All specifications include school fixed effects and baseline controls (repeater status, class size, incentive condition). Delta-method standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table S22. Replication of Table 1 using multiple imputation by chained equations (MICE-PMM, $M = 20$ datasets).

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.019**	0.019**	0.022**	0.019**	0.024***	0.024***	0.031***	0.024***
	(0.009)	(0.009)	(0.010)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
CogAb(r)	0.073***	0.101*	0.072***	0.072***	0.073***	0.092	0.072***	0.074***
	(0.011)	(0.061)	(0.011)	(0.011)	(0.010)	(0.057)	(0.010)	(0.010)
Female	−0.045**	−0.046**	0.035	−0.046**	−0.052***	−0.052***	0.135	−0.050**
	(0.021)	(0.021)	(0.108)	(0.021)	(0.020)	(0.020)	(0.101)	(0.020)
Age × CogAb(r)		−0.002				−0.001		
		(0.004)				(0.004)		
Age × Female			−0.006				−0.013*	
			(0.007)				(0.007)	
CogAb(r)^2				−0.005				0.011
				(0.009)				(0.008)
Constant	0.573***	0.575***	0.536***	0.579***	0.121	0.122	0.033	0.109
	(0.147)	(0.147)	(0.155)	(0.148)	(0.139)	(0.139)	(0.146)	(0.139)
Observations	2,131	2,131	2,131	2,131	2,118	2,118	2,118	2,118
Imputation	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM	MICE-PMM
M	20	20	20	20	20	20	20	20
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Note: Replication of Table 1 replacing the CRT rescaling for Study 2 participants with multiple imputation by chained equations (predictive mean matching, $M = 20$ imputed datasets). Estimates and standard errors combine the imputed datasets using Rubin's rules. In columns 2 and 6 the CogAb(r) coefficient is evaluated at age zero and is therefore not comparable to the other columns. All specifications include school fixed effects and baseline controls (repeater status, class size, incentive condition). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Data and Code Availability. The data and replication code supporting the findings of this study are publicly available in the Zenodo repository (https://doi.org/10.5281/zenodo.19610263).

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